RETAIL SALES ANALYSIS

SQL Project Report

1. Project Overview

This project uses PostgreSQL to explore retail transaction data and answer practical business questions. The analysis focuses on sales performance, product categories, customer purchasing patterns, monthly revenue, and transaction timing. SQL is used to inspect the dataset, clean selected data-quality issues, summarize records, and produce results that can support business discussions.

Project Objectives

- Inspect the structure and completeness of the retail sales dataset.
- Standardize gender values and identify records with missing information.
- Measure sales and transaction activity by product category and customer group.
- Identify high-value transactions, top-spending customers, and best-selling months.
- Summarize transaction activity by time-of-day shift.

2. Dataset Description

The retail_sales table is designed to store transaction-level retail data. Each row represents a recorded sales transaction, identified by transaction_id. The dataset includes transaction date and time, customer information, product category, quantity, unit price, cost of goods sold, and total sale amount.

- **transaction_id:** Integer identifier for each transaction; defined as the primary key.
- **sale_date:** Date on which the transaction took place.
- **sale_time:** Time at which the transaction took place.
- **customer_id:** Identifier for the customer associated with the transaction.
- **gender:** Customer gender value recorded in the source data; M/F values are standardized to Male/Female.
- **age:** Customer age, where available.
- **category:** Product category, such as Clothing or Beauty.
- **price_per_unit:** Price charged for one unit of the product.
- **quantity:** Number of units recorded in the transaction.
- **cogs:** Cost of goods sold associated with the transaction.
- **total_sale:** Total sale amount recorded for the transaction.

Data note: the table definition creates the structure only. The source dataset must be imported or inserted before the cleaning and analysis queries will return meaningful results. The SQL supplied for this report does not include a row count, sample records, or query outputs, so no numerical findings are claimed here.

3. Table Creation and Initial Inspection

The script drops an existing retail_sales table and recreates it. This is useful when rebuilding a practice project, but DROP TABLE permanently removes the table and its data. Do not run it against a production table unless data loss is intended.

```
-- Create the table
DROP TABLE IF EXISTS retail_sales;

CREATE TABLE retail_sales (
    transaction_id INT PRIMARY KEY,
    sale_date DATE,
    sale_time TIME,
    customer_id INT,
    gender VARCHAR(15),
    age INT,
    category VARCHAR(15),
    price_per_unit FLOAT
    quantity INT,
    cogs FLOAT
    total_sale FLOAT
);

-- View the records
SELECT *
FROM retail_sales;
```

4. Data Cleaning

The cleaning stage checks for missing values, removes rows with NULL values in the selected fields, and standardizes gender labels. The following queries include blank-text checks for gender and category as well as trimming surrounding spaces.

4.1 Find rows with missing values

```
SELECT *
FROM retail_sales
WHERE transaction_id IS NULL
   OR sale_date IS NULL
   OR sale_time IS NULL
   OR customer_id IS NULL
   OR gender IS NULL
   OR age IS NULL
   OR category IS NULL
   OR quantity IS NULL
   OR price_per_unit IS NULL
   OR cogs IS NULL
   OR total_sale IS NULL;
```

4.2 Delete rows with missing values or blank text fields

```
DELETE FROM retail_sales
WHERE transaction_id IS NULL
   OR sale_date IS NULL
   OR sale_time IS NULL
   OR customer_id IS NULL
   OR gender IS NULL
   OR age IS NULL
   OR category IS NULL
   OR quantity IS NULL
   OR price_per_unit IS NULL
   OR cogs IS NULL
   OR total_sale IS NULL;
```

4.3 Standardize gender values

```
SELECT DISTINCT gender
FROM retail_sales;

UPDATE retail_sales
SET gender='Male'
WHERE gender='M';
```

```
UPDATE retail_sales
SET gender='Female'
WHERE gender='F';
```

4.4 Verify cleaned data

```
SELECT *
FROM retail_sales;
```

5. Business Questions and SQL Analysis

The queries below address the ten business questions defined for this project. Each result should be interpreted using the actual query output from the imported dataset.

Question 1. Sales on November 5, 2022

Business purpose: retrieve all transaction records for a specific date.

```
SELECT *
FROM retail_sales
WHERE sale_date = DATE '2022-11-05';
```

	transaction_id [PK] integer	sale_date date	sale_time time without time zone	customer_id integer	gender character varying (15)	age integer	category character varying (15)	price_per_unit double precision	quantity integer	cogs double precision	total_sale double precision
1	696	2022-11-05	10:47:00	117	Female	41	Clothing	300	3	129	900
2	697	2022-11-05	11:49:00	95	Female	23	Beauty	300	1	123	300
3	698	2022-11-05	14:35:00	86	Female	55	Clothing	300	3	111	900
4	699	2022-11-05	16:31:00	53	Female	20	Beauty	30	2	8.1	60
5	700	2022-11-05	17:43:00	102	Female	54	Electronics	30	4	9.3	120
6	701	2022-11-05	19:29:00	90	Male	57	Clothing	300	4	318	1200
7	702	2022-11-05	20:06:00	140	Female	40	Beauty	300	4	105	1200
8	703	2022-11-05	20:19:00	87	Male	30	Electronics	25	2	30.75	50
9	704	2022-11-05	20:44:00	83	Female	35	Beauty	50	2	13.5	100
10	705	2022-11-05	22:34:00	104	Female	46	Beauty	500	2	145	1000
11	695	2022-11-05	09:58:00	29	Male	23	Clothing	500	2	190	1000

Total rows: 11 | Query complete 00:00:00.135 | CRLF | Ln 117, Col 15

Interpretation: The output contains all available columns for transactions recorded on November 5, 2022.

Question 2. Clothing transactions with quantity greater than 2 in November 2022

Business purpose: examine larger-quantity Clothing transactions during November 2022.

```
SELECT *
FROM retail_sales
WHERE category = 'Clothing'
  AND quantity > 2
  AND sale_date >= DATE '2022-11-01'
  AND sale_date < DATE '2022-12-01'
ORDER BY quantity;
```

	transaction_id [PK] integer	sale_date date	sale_time time without time zone	customer_id integer	gender character varying (15)	age integer	category character varying (15)	price_per_unit double precision	quantity integer	cogs double precision	total_sale double precision
1	750	2022-11-13	14:48:00	52	Female	51	Clothing	50	3	24.5	150
2	696	2022-11-05	10:47:00	117	Female	41	Clothing	300	3	129	900
3	793	2022-11-22	18:30:00	86	Male	22	Clothing	300	3	300	900
4	758	2022-11-14	14:44:00	104	Female	31	Clothing	500	3	200	1500
5	800	2022-11-24	12:12:00	80	Male	55	Clothing	300	3	132	900
6	698	2022-11-05	14:35:00	86	Female	55	Clothing	300	3	111	900
7	748	2022-11-13	07:33:00	69	Female	35	Clothing	25	3	8.75	75
8	820	2022-11-29	17:43:00	46	Female	35	Clothing	50	3	20.5	150
9	776	2022-11-19	20:44:00	107	Male	58	Clothing	500	3	130	1500
10	707	2022-11-06	19:21:00	64	Male	39	Clothing	25	3	27.5	75
11	771	2022-11-18	17:28:00	149	Male	27	Clothing	300	3	99	900

Total rows: 29 | Query complete 00:00:00.176 | CRLF | Ln 125, Col 15

Interpretation: The query filters by category, quantity, and date range, then sorts the matching records by quantity in ascending order.

Question 3. Sales revenue and transaction count by category

Business purpose: compare the sales contribution and recorded transaction count for each product category.

```
SELECT
    category,
    SUM(total_sale) AS net_sales,
    COUNT(transaction_id) AS total_orders
FROM retail_sales
GROUP BY category
ORDER BY net_sales DESC;
```

	category character varying (15)	net_sales double precision	total_orders bigint
1	Electronics	313810	684
2	Clothing	311070	701
3	Beauty	286840	612

Interpretation: Each category receives a total sales amount and a count of transaction IDs. This count represents rows/transaction IDs, not necessarily distinct orders if an order can span multiple rows.

Question 4. Average age of Beauty customers

Business purpose: calculate the average recorded age among customers associated with Beauty transactions.

```
SELECT ROUND(AVG(age), 2) AS avg_customer_age
FROM retail_sales
WHERE category = 'Beauty';
```

	avg_customer_age numeric
1	40.39

Interpretation: AVG ignores NULL age values. The result is rounded to two decimal places.

Question 5. Transactions with total sales greater than 1,000

Business purpose: identify high-value transactions above the 1,000 threshold.

```
SELECT *
FROM retail_sales
WHERE total_sale >= 1000
ORDER BY total_sale DESC;
```

	transaction_id [PK] integer	sale_date date	sale_time time without time zone	customer_id integer	gender character varying (15)	age integer	category character varying (15)	price_per_unit double precision	quantity integer	cogs double precision	total_sale double precision
1	495	2022-09-22	22:32:00	61	Male	19	Electronics	500	2	140	1000
2	505	2022-09-24	12:30:00	91	Female	63	Electronics	500	2	265	1000
3	1389	2023-08-04	11:38:00	8	Female	19	Electronics	500	2	230	1000
4	944	2022-12-25	06:24:00	67	Female	26	Electronics	500	2	245	1000
5	938	2022-12-24	16:38:00	73	Female	51	Beauty	500	2	125	1000
6	913	2022-12-18	17:44:00	67	Female	59	Beauty	500	2	575	1000
7	839	2022-12-03	12:40:00	100	Female	48	Beauty	500	2	145	1000
8	491	2022-09-22	10:49:00	2	Female	40	Beauty	500	2	160	1000
9	808	2022-11-26	18:17:00	112	Male	51	Beauty	500	2	555	1000
10	1161	2023-04-14	07:12:00	4	Male	23	Clothing	500	2	150	1000
11	564	2022-10-08	17:24:00	56	Female	22	Beauty	500	2	580	1000

Total rows: 404 | Query complete 00:00:00.104 | CRLF | Ln 161, Col 11

Interpretation: The query returns transactions strictly greater than 1,000 and lists the largest amounts first. The greater-than sign matches the wording “exceeded 1,000.”

Question 6. Transaction count by gender and category

Business purpose: compare transaction activity across gender and product category.

```
SELECT
    gender,
    category,
    COUNT(transaction_id) AS total_transactions
FROM retail_sales
GROUP BY gender, category
ORDER BY category, gender;
```

	gender character varying (15)	category character varying (15)	total_transactions bigint
1	Female	Beauty	316
2	Female	Clothing	348
3	Female	Electronics	354
4	Male	Electronics	330
5	Male	Clothing	353
6	Male	Beauty	296

Interpretation: The results show one row for each gender/category combination and the corresponding transaction ID count.

Question 7. Best-selling month in each year

Business purpose: aggregate monthly sales and identify the month or months with the highest total sales within each year.

```
WITH monthly_sales AS (
     SELECT
             EXTRACT(YEAR FROM sale_date) AS year,
             EXTRACT(MONTH FROM sale_date) AS month,
             SUM(total_sale) AS total_sales,
             RANK() OVER( PARTITION BY EXTRACT(YEAR FROM sale_date) ORDER BY SUM(total_sale) DESC
                                 ) AS rnk
                                 FROM retail_sales
                                 GROUP BY 1,2
                                 )
                                 SELECT * FROM
                                 monthly_sales
                                 WHERE rnk=1;
```

	year numeric	month numeric	total_sales double precision	rnk bigint
1	2022	12	72880	1
2	2023	12	69145	1

Interpretation: The CTE first calculates monthly revenue. RANK then ranks months within each year. If months tie for the highest sales, all tied months are returned.

Question 8. Top five customers by total sales

Business purpose: identify the five customers associated with the highest cumulative sales amount.

```
SELECT
    customer_id,
    SUM(total_sale) AS total_sales
FROM retail_sales
GROUP BY customer_id
ORDER BY total_sales DESC
LIMIT 5;
```

	customer_id integer	total_sales double precision
1	3	38440
2	1	30750
3	5	30405
4	2	25295
5	4	23580

Interpretation: The output ranks customers by the sum of their recorded transaction sales and returns up to five customers.

Question 9. Unique customers by category

Business purpose: understand how many distinct customers purchased from each product category.

```
SELECT
    category,
    COUNT(DISTINCT customer_id) AS unique_customers
FROM retail_sales
GROUP BY category
ORDER BY unique_customers DESC;
```

	category character varying (15)	unique_customers bigint
1	Clothing	149
2	Electronics	144
3	Beauty	141

Interpretation: COUNT(DISTINCT customer_id) counts each customer once per category, even if that customer has multiple transactions in the category.

Question 10. Transactions by time-of-day shift

Business purpose: summarize when transactions occur using the defined Morning, Afternoon, and Evening time ranges.

```
SELECT
    CASE
        WHEN sale_time < TIME '12:00:00' THEN 'MORNING'
        WHEN sale_time < TIME '18:00:00' THEN 'AFTERNOON'
        ELSE 'EVENING'
    END AS shift,
    COUNT(*) AS total_orders
FROM retail_sales
GROUP BY shift
ORDER BY total_orders DESC;
```

	shift text	total_orders bigint
1	EVENING	1062
2	MORNING	558
3	AFTERNO...	377

Interpretation: Morning means before 12:00 PM, Afternoon means 12:00 PM to before 6:00 PM, and Evening means 6:00 PM onward. COUNT(*) counts rows; it counts unique orders only if each row represents one order.

6. Additional Validation

6.1 Total records

```
SELECT COUNT(*) AS total_records
FROM retail_sales;
```

6.2 Overall sales

```
SELECT SUM(total_sale) AS overall_sales
FROM retail_sales;
```

6.3 Check duplicate transaction IDs

```
SELECT transaction_id, COUNT(*) AS duplicate_count
FROM retail_sales
GROUP BY transaction_id
HAVING COUNT(*) > 1;
```

With the primary key constraint enabled, duplicate transaction_id values cannot be stored in this table. This check is mainly useful when validating a source file before importing it or when the source table does not enforce a primary key.

7. Key Findings and Business Value

7.1 key Findings

1. Sales Performance by Category: Electronics generated the highest total sales of 313,810, followed by Clothing at 311,070 and Beauty at 286,840. Electronics performed slightly better than the other categories in total sales.
2. Beauty Customer Demographics: The average age of customers purchasing Beauty products was 40.39 years. This helps describe the customer demographic for this category.
3. Top-Spending Customers: Customer ID 3 recorded the highest total purchases at 38,440, followed by Customer ID 1 at 30,750 and Customer ID 5 at 30,405. These customers may be useful segments for customer-retention analysis.
4. Monthly Sales Trends: December was the highest-sales month in both years shown in the query results. Sales were 72,880 in December 2022 and 69,145 in December 2023. This suggests a recurring year-end sales pattern in the available data.
5. Sales by Time of Day: The evening shift recorded 1,062 transactions, followed by the morning shift with 558 and the afternoon shift with 377. Evening had the highest transaction count among the three time periods.
6. Unique Customers by Category: Clothing had 149 unique customers, Electronics had 144, and Beauty had 141. Clothing reached the largest number of unique customers among the three categories.
7. Gender-Based Shopping Patterns: The query results show differences in transaction counts by gender and product category. For example, female customers recorded 354 Electronics transactions, while male customers recorded 353 Clothing transactions. These results can help guide further analysis of purchasing patterns.

7.2 Business Value

- Better Sales Decisions: Comparing category sales helps management understand which product categories contribute most to total revenue.
- Improved Marketing: Customer demographics and category-level shopping patterns can help businesses develop more relevant marketing campaigns.
- Customer Retention: Identifying high-spending customers can support loyalty programs and targeted offers.
- Seasonal Planning: December's high sales in both years suggest that the business could investigate year-end promotions and prepare inventory accordingly.
- Operational Planning: The high evening transaction count can help management assess staffing and checkout requirements during busy periods.
- Customer Reach: Comparing unique customers across categories helps identify opportunities to attract more customers to lower-reach categories.
- Data-Driven Decisions: SQL makes it possible to organize, filter, aggregate, and analyze sales data to support business decisions.

8. Conclusion

The analysis provides useful insights into product category performance, customer demographics, high-spending customers, monthly sales trends, and transaction patterns throughout the day. These findings can support better marketing, inventory, and operational planning. Further analysis would be needed to confirm the reasons behind these patterns and determine their impact on profitability.